SecureCorp Home Lab

Cybersecurity Lab Project Report

Report status	Scope
Submission-ready report based on verified available evidence	Home laboratory documentation and assessment

Purpose

This reportpresents the SecureCorp Home Lab as a controlled cybersecurity learning environment. It records the lab components and verified evidence available at the time of preparation, defines the project requirements, and distinguishes completed work from activities that remain unverified.

Evidence integrity statement

The available record confirms Nmap enumeration and custom Wazuh rules 100100 and 100101. It does not provide raw scan output, screenshots, alert records, exploit results, timestamps, target addresses, or remediation validation. Those items are not claimed as completed in this report.

Prepared for project submission

Date: 17 October 2026

1 Project overview

SecureCorp Home Lab isaself-contained environment for practising security monitoring, service discovery, controlled testing, and investigation. The documented stack combines an attacker or analyst workstation, intentionally vulnerable targets, and a SIEM platform so that technical activity can be observed and evaluated without representing unverified outcomes as facts.

Objectives

- Build a controlled environment for network and host-security learning.
- Use Nmap for authorised service enumeration within the lab.
- Collect and review security telemetry through Wazuh.
- Develop and maintain custom Wazuh detection rules.
- Document evidence, investigation steps, limitations, and next actions honestly.

Lab components

Component	Role in the lab	Status in available record
Kali Linux / WSL2	Security testing and analyst workstation.	Documented component
Metasploitable2	Intentionally vulnerable target for isolated exercises.	Documented component
DVWA	Deliberately vulnerable web application for authorised practice.	Documented component
Wazuh	SIEM and monitoring platform for collection, alerting, and investigation.	Documented component
Nmap	Network and service enumeration tool.	Verified use recorded

Safety boundary

All testing described or planned for this project is limited to systems owned, configured, or explicitly authorised for the home lab. The report does not authorise testing of public systems, third-party networks, or services outside this boundary.

2 Project requirements

The followingrequirements arederived from the project description and repository review recorded in the referenced conversation. A requirement is not treated as completed unless supporting evidence is explicitly listed in Section 3.

ID	Requirement	Current evidence state
R1	Provide a lab with Kali/WSL2, Metasploitable2, DVWA, Wazuh, and Nmap.	Components documented; configuration evidence unavailable
R2	Create and maintain a network architecture description.	Requirement documented; diagram unavailable

ID	Requirement	Current evidence state
R3	Perform authorised Nmap enumeration in the lab.	Verified as performed; raw results unavailable
R4	Configure Wazuh SIEM monitoring and custom detection rules.	Rules 100100 and 100101 verified as documented
R5	Document detection and investigation workflow.	Requirement documented; case evidence unavailable
R6	Conduct controlled security testing only when supported by evidence.	No attack outcome verified
R7	Preserve an evidence-based lab status and avoid claiming unverified activity.	Verified principle recorded in LAB-STATUS documentation

3 Verified evidence and findings

Thissection usesadeliberatelynarrow definition ofverified: a result is marked verified only where the referenced repository review explicitly recorded it.

Evidence item	Verified finding	What cannot be concluded
Nmap enumeration	An Nmap enumeration activity was recorded as verified in the lab status.	Open ports, services, versions, hosts, dates, and scan command details are not available.
Wazuh custom rules	Rules 100100 and 100101 were recorded as present in the lab status.	Their exact rule logic, enabled state, triggering events, and alert output are not available.
Evidence discipline	The lab status explicitly distinguishes unverified attack results from completed work.	This does not prove any specific attack, detection, or remediation exercise occurred.

Assessment

The verified evidence supports describing SecureCorp as a documented home-lab project with established discovery activity and custom Wazuh rule work. It does not yet support quantitative claims about detection coverage, exploited vulnerabilities, incident timelines, alert fidelity, or remediation effectiveness.

4 Architecture and workflow

The intended workflow isaclosed-looplearning process. It should be represented as an operational design, not as proof that every stage has already been completed.

Stage	Intended activity	Evidence needed to mark complete
Discover	Enumerate lab hosts and services with Nmap.	Command, date, scope, sanitised output, and service inventory.
Test	Run authorised exercises against DVWA or other owned lab targets.	Test plan, scope confirmation, reproducible steps, and artefacts.

Stage	Intended activity	Evidence needed to mark complete
Detect	Collect events in Wazuh and evaluate relevant rules.	Rule source, alert ID, timestamp, event context, and screenshot or export.
Investigate	Triage an alert and reconstruct relevant activity.	Timeline, log sources, analyst notes, and conclusion.
Remediate	Apply a safe hardening change and retest.	Before/after configuration, validation output, and rollback note.

Architecture note

A network diagram is referenced as a project artefact but was not available for inspection. Any future diagram should identify the analyst system, targets, Wazuh components, network segmentation, and permitted traffic paths without publishing sensitive addresses or credentials.

5 Unverified activities and evidence gap

The following items may be planned or described as part of the lab, but they are not verified by the available source material and therefore are not reported as completed results.

- DVWA vulnerability testing, including authentication testing, SQL injection, cross-site scripting, or command injection.
- Metasploitable2 exploitation, FTP/SSH testing, credential testing, or post-exploitation activity.
- End-to-end Wazuh alert generation and correlation for a specific exercise.
- Incident-response timelines, root-cause analysis, containment, or recovery outcomes.
- Hardening changes such as firewall configuration, SSH hardening, service disablement, or account controls.
- Measured detection rates, false-positive rates, response times, or vulnerability counts.

To convert any of these into a completed submission claim, retain reproducible evidence: approved scope, commands or procedures, timestamps, raw or sanitised outputs, screenshots, rule/configuration snapshots, and a short interpretation of the result.

6 Recommended evidence package

The next submission revision should add a small, auditable evidence set rather than broad unsupported claims.

Folder or artefact	Minimum contents
architecture/network-diagram.png	Segmented topology, roles, data flows, and lab-only boundary.
discovery/nmap-scan.txt	Sanitised output, command, date, and target scope.
discovery/service-inventory.md	Host, service, port, exposure rationale, and owner.
detection/wazuh-rules.xml	Rules 100100/100101 source and concise rationale.

Folder or artefact	Minimum contents
detection/screenshots	Alert views with time and identifying details appropriately sanitised.
incidents/incident-001.md	Trigger, evidence, timeline, impact assessment, and decision log.
remediation/hardening.md	Change record, validation, residual risk, and rollback.

7 Conclusion

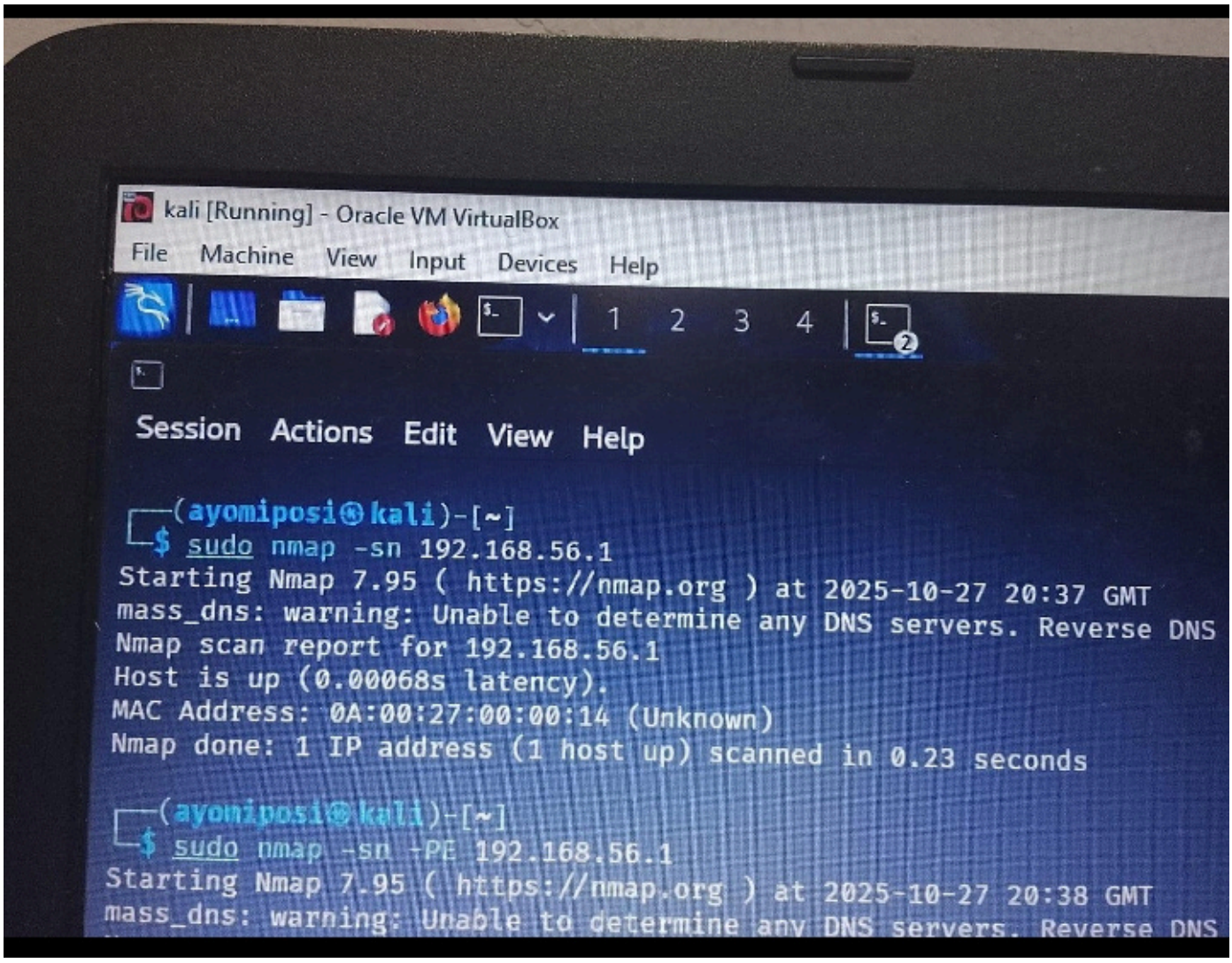
SecureCorpHome Lab is a credible cybersecurity portfolio foundation because it combines a controlled practice environment, service discovery, SIEM monitoring, and custom Wazuh rule work. The evidence currently available verifies Nmap enumeration and the presence of Wazuh rules 100100 and 100101, while preserving the distinction between documented design and tested outcomes. The highest-value next step is to add compact, reproducible evidence for one authorised exercise through discovery, detection, investigation, and remediation.

Source and limitation

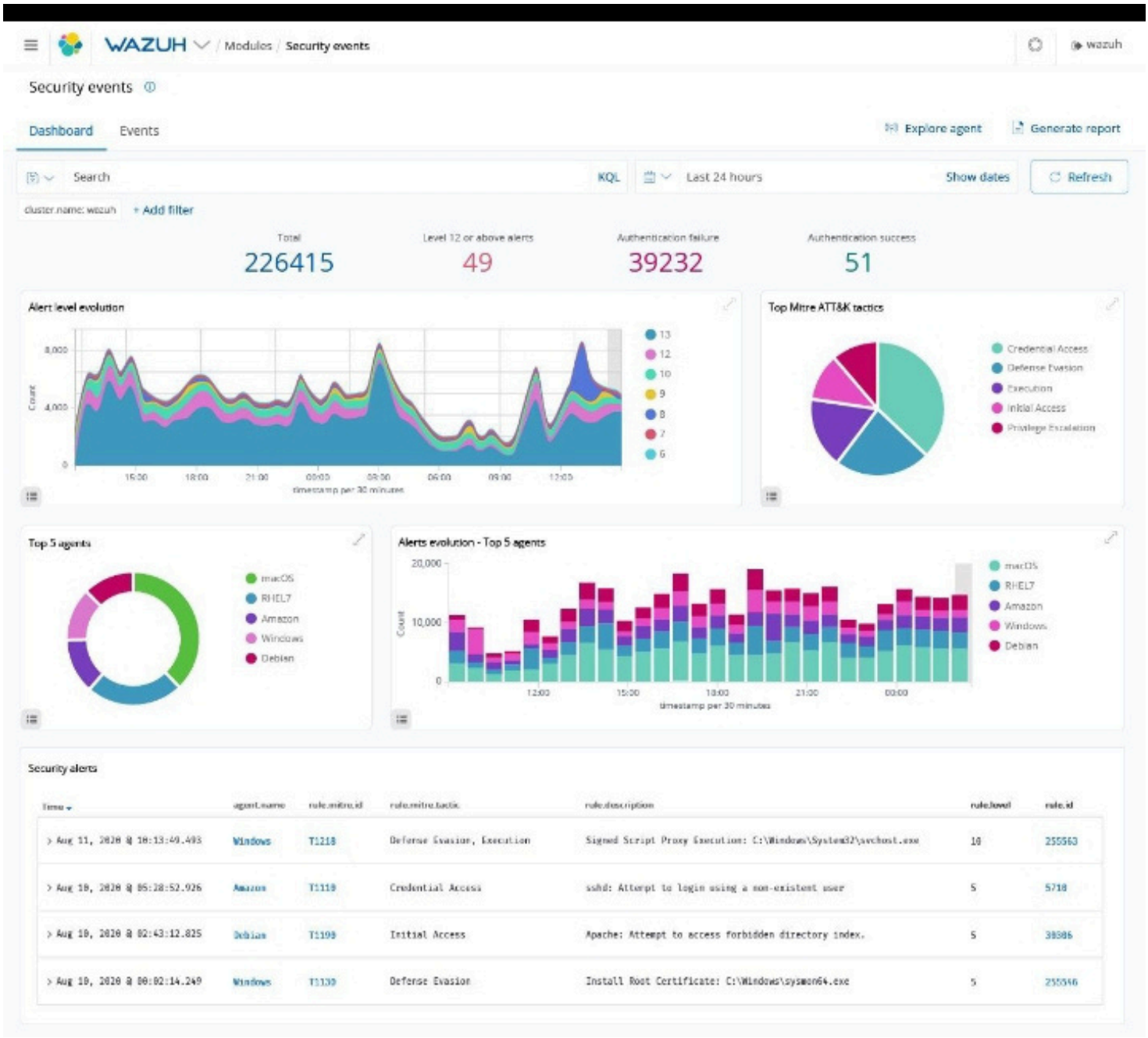
This report was prepared from the available referenced conversation because no prior PDF, Home_lab_setup.docx, repository checkout, raw evidence, or attachments were present in the accessible workspace. It should be updated when those primary artefacts are available.

Screenshot proofs

Nmap -scan.png



Wazuh dashboard



DVWA security

